

# Mandatory Assignment 1

GFS189, RTL959

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In this assignment we study estimation error in mean-variance portfolio optimization, replicating the main findings of [Jobson and Korkie \(1980\)](#) and [Brandt \(2010\)](#). We calibrate a single-factor model from S&P 500 data, construct a synthetic universe of 50 assets with known true parameters, and use Monte Carlo simulation to evaluate the out-of-sample performance of several portfolio strategies.

We study how plug-in portfolio weights  $\hat{\omega}(\bar{\mu})$  deviate from the theoretically optimal  $\omega(\bar{\mu})$  when sample estimates replace true parameters. The plug-in problem is:

$$\hat{\omega}(\bar{\mu}) = \arg \min_{\omega} \omega' \hat{\Sigma} \omega \quad \text{s.t.} \quad \omega' \iota = 1 \quad \text{and} \quad \omega' \hat{\mu} \geq \bar{\mu}$$

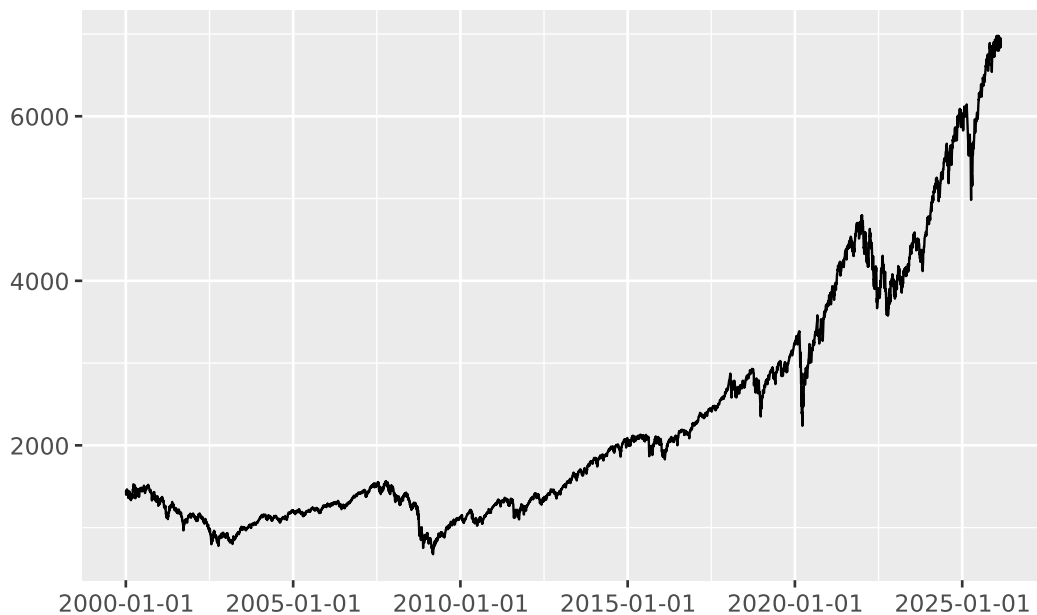
## 1 Synthetic Asset Universe

We construct a universe of  $N = 50$  synthetic assets following the single-factor model:

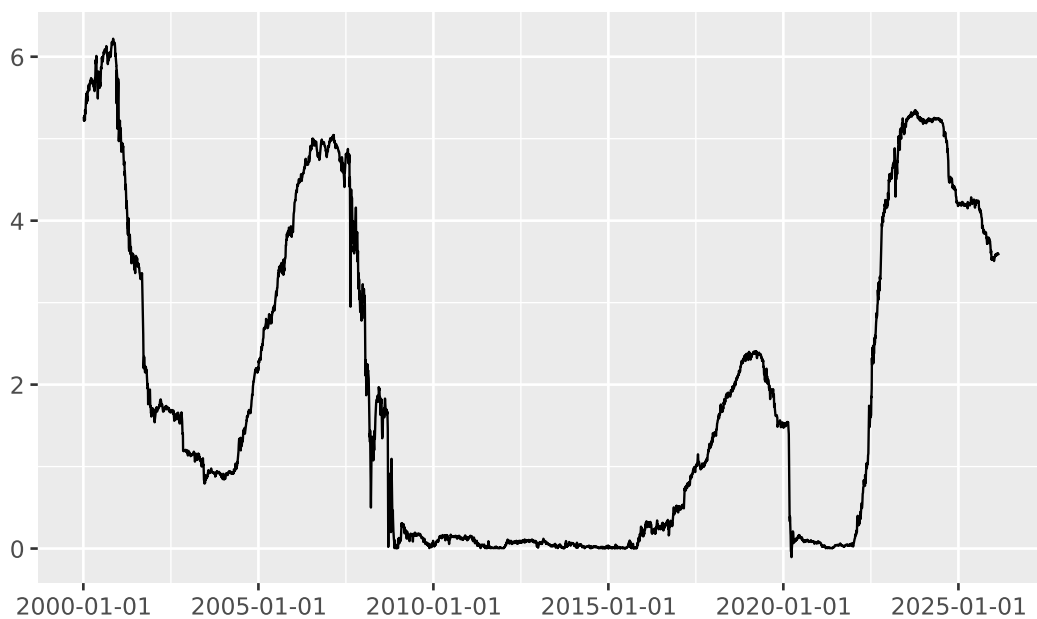
$$r_{i,t} = \alpha_i + \beta_i r_{m,t} + \epsilon_{i,t}$$

where  $r_{m,t}$  is the market return,  $\alpha_i$  and  $\beta_i$  are the intercept and factor loading for asset  $i$ , and  $\epsilon_{i,t}$  is the idiosyncratic shock.

S&P 500 stock prices (in USD) 2000-2026



13-week T-Bill (in USD) 2000-2026



We now use the data to calculate these: Let  $r_{m,t}$  be the monthly return and  $r_{f,t}$  the risk-free rate.

### 1. Monthly Excess Return

$$\tilde{r}_{m,t} = r_{m,t} - r_{f,t}$$

### 2. Monthly Mean Excess Return

$$\hat{\mu}_m = \frac{1}{T} \sum_{t=1}^T \tilde{r}_{m,t}$$

### 3. Monthly Variance

$$\hat{\sigma}_m^2 = \frac{1}{T-1} \sum_{t=1}^T (\tilde{r}_{m,t} - \hat{\mu}_m)^2$$

### 4. Annualized Mean Excess Return

$$\hat{\mu}_{m,\text{ann}} = 12 \times \hat{\mu}_m$$

### 5. Annualized Variance

$$\hat{\sigma}_{m,\text{ann}}^2 = 12 \times \hat{\sigma}_m^2$$

|                                |       |
|--------------------------------|-------|
| Monthly mean excess return:    | 0.006 |
| Monthly variance:              | 0.002 |
| Annualized mean excess return: | 0.071 |
| Annualized variance:           | 0.023 |

The annualized mean excess return and variance are reported above. The mean excess return is positive over the sample period, consistent with the equity risk premium.

## 2 Synthetic Asset Universe

We construct a universe of  $N = 50$  synthetic assets following the single-factor model: **Single Asset Return Process**

$$\tilde{r}_{i,t} = \beta_i \tilde{r}_{m,t} + \varepsilon_{i,t}, \quad \varepsilon_{i,t} \stackrel{iid}{\sim} \mathcal{N}(0, \sigma_{\varepsilon,i}^2), \quad i = 1, \dots, N$$

where  $\tilde{r}_{i,t}$  denotes the excess return on asset  $i$ .

### Parameter Draws

$$\beta_i \stackrel{iid}{\sim} \text{Uniform}(0.5, 1.5), \quad \sigma_{\varepsilon,i} \stackrel{iid}{\sim} \text{Uniform}\left(\frac{0.10}{\sqrt{12}}, \frac{0.30}{\sqrt{12}}\right)$$

### True Monthly Parameters

$$\mu_i = \beta_i \hat{\mu}_m, \quad \Sigma = \sigma_m^2 \beta \beta' + \text{diag}(\sigma_{\varepsilon}^2)$$

where  $\beta = (\beta_1, \dots, \beta_N)'$ .

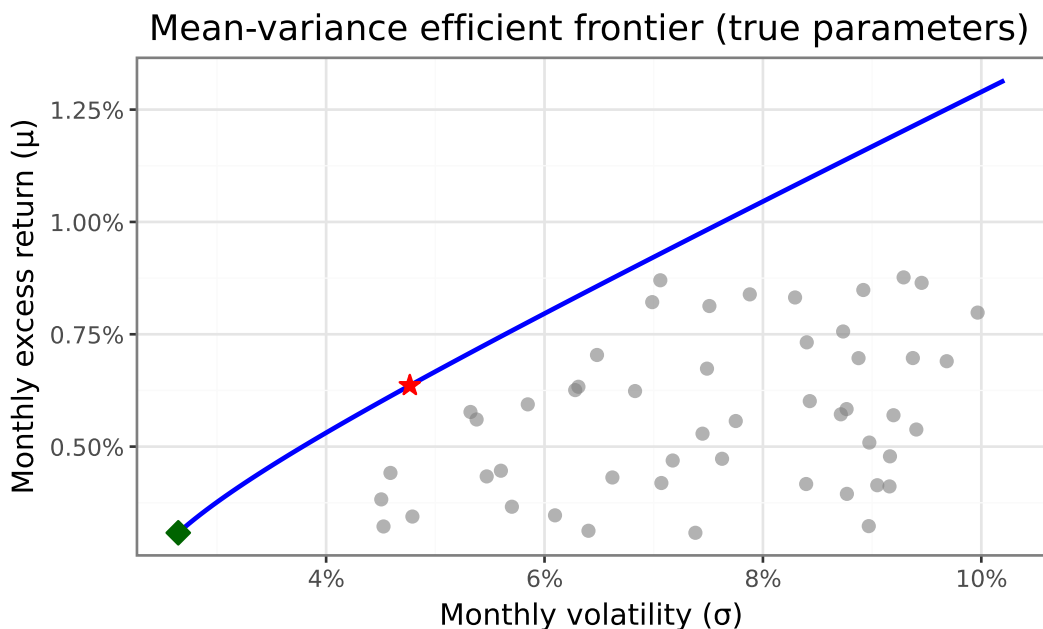
The resulting parameter matrix  $\Sigma$  is positive definite by construction, ensuring the frontier is well-defined. All 50 assets have positive expected excess returns (inherited from  $\hat{\mu}_m > 0$ ), with cross-asset correlations driven entirely by their common factor exposure.

### 3 Efficient Frontier Function

We implement a reusable function to compute the mean-variance efficient frontier and the tangency portfolio analytically using the two-fund separation theorem.

### 4 Efficient Frontier Visualization

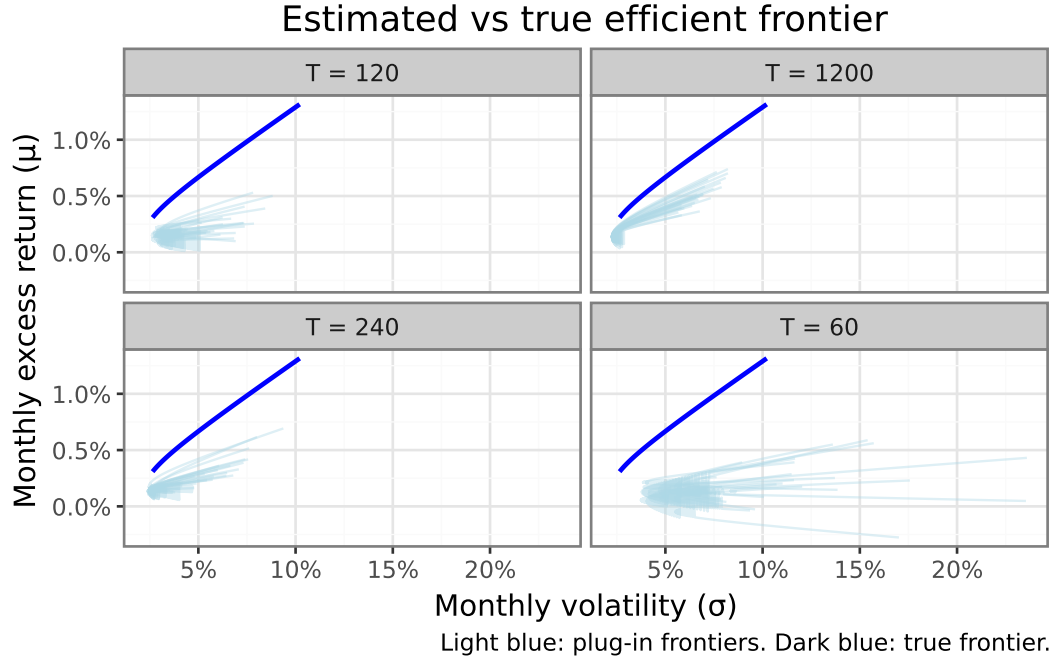
The figure below plots the mean-variance efficient frontier constructed from the true parameters  $\mu$  and  $\Sigma$ . Individual assets (grey dots) lie below the frontier, confirming that diversification dominates holding any single asset. The red star marks the tangency portfolio (maximum Sharpe ratio) and the green diamond marks the minimum variance portfolio.



gency portfolio. Green diamond: minimum variance portfolio. Grey dots: individual assets.

### 5 Estimation Error and the Efficient Frontier

We now show how estimation error distorts the shape of the efficient frontier. For each sample size  $T \in \{60, 120, 240, 1200\}$  (corresponding to 5, 10, 20, and 100 years of monthly data), we draw 25 synthetic samples from the true DGP, estimate plug-in parameters, and plot the resulting frontier evaluated at the true parameters.

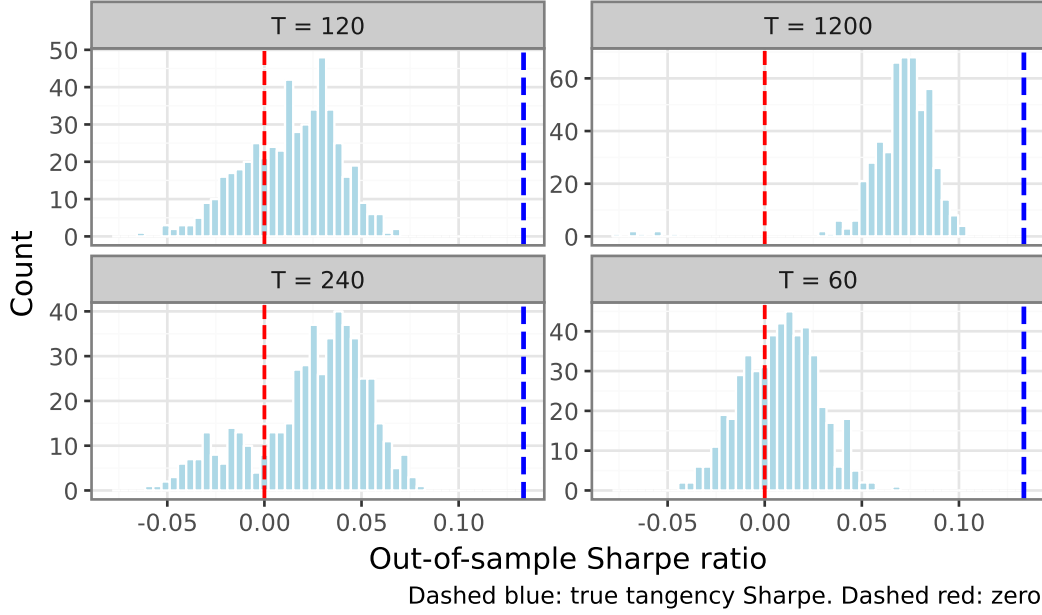


As  $T$  increases, the plug-in frontiers converge toward the true frontier, though substantial dispersion persists even at  $T = 1200$  (100 years of monthly data). The severity of estimation error is governed by three factors: (1) the ratio  $N/T$ , a large number of assets relative to observations degrades the precision of both  $\hat{\mu}$  and  $\hat{\Sigma}$ ; (2) sample size in absolute terms, larger  $T$  reduces sampling variance; and (3) asset correlation, higher correlation makes the covariance matrix ill-conditioned, amplifying inversion errors. These findings motivate shrinkage estimators and structured factor models as practical alternatives to plug-in estimation.

## 6 Out-of-Sample Performance of the Plug-In Tangency Portfolio

We evaluate the out-of-sample Sharpe ratio of the plug-in tangency portfolio across 500 simulations for each  $T$ . Weights are computed from estimated parameters but performance is measured using the true  $\mu$  and  $\Sigma$ , isolating the cost of estimation error. The Sharpe ratio is used as the performance criterion, as it captures the trade-off between return and risk on a scale comparable across strategies.

## Out-of-sample Sharpe ratio of plug-in tangency portfolio

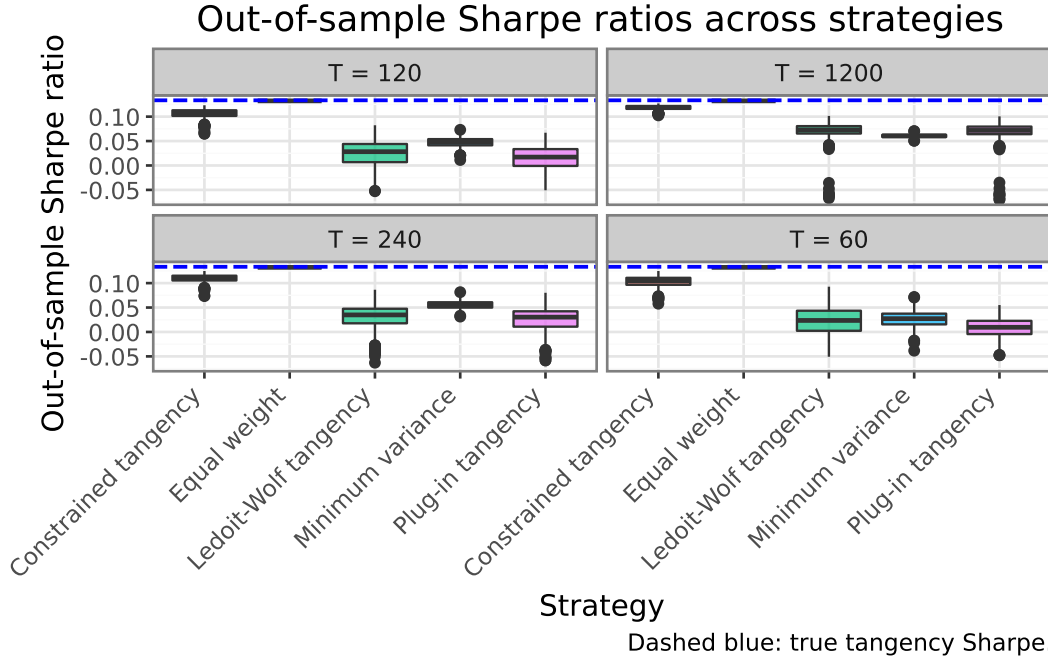


Across all sample sizes, the plug-in tangency portfolio performs below the true optimum (dashed blue line). As  $T$  increases, the distribution of out-of-sample Sharpe ratios shifts right and narrows, confirming that larger samples reduce but do not eliminate estimation error. At  $T = 60$ , a substantial share of simulations produces negative Sharpe ratios, highlighting the practical cost of estimation error when data are scarce.

## 7 Alternative Portfolio Strategies

We compare four strategies against the equal-weight benchmark ( $\omega_{ew} = \frac{1}{N}\iota$ ) across 500 simulations for  $T \in \{60, 120, 240, 1200\}$ , always evaluating performance with the true  $\mu$  and  $\Sigma$ :

- **Plug-in tangency:** the unconstrained mean-variance optimum using  $\hat{\mu}$  and  $\hat{\Sigma}$ . It is the theoretically correct estimator but amplifies estimation error freely.
- **Minimum variance portfolio:** minimizes  $\omega' \hat{\Sigma} \omega$  subject to  $\omega' \iota = 1$ , ignoring expected returns entirely. Since  $\hat{\mu}$  is typically the noisiest input in finite samples, avoiding it altogether reduces the main source of estimation error.
- **Short-sell constrained tangency:** imposes  $\omega_i \geq 0$  on the Sharpe-maximizing problem. The constraint prevents extreme long/short positions driven by sampling noise, trading some theoretical optimality for robustness.
- **Ledoit-Wolf shrinkage tangency:** replaces  $\hat{\Sigma}$  with the Ledoit-Wolf shrinkage estimator, which pulls extreme sample eigenvalues toward a structured target. This stabilizes the covariance inverse that drives portfolio weights, directly attacking the source of estimation error in  $\Sigma$ .



The figure shows that the equal-weight portfolio achieves the highest and most stable Sharpe ratio across all sample sizes, confirming that naive diversification is difficult to beat out-of-sample, a result consistent with [Jobson and Korkie \(1980\)](#). The Ledoit-Wolf shrinkage tangency portfolio outperforms the unconstrained plug-in tangency at small  $T$ , demonstrating that regularizing the covariance estimate directly reduces the most damaging source of estimation error. The short-sell constrained tangency performs similarly to Ledoit-Wolf at small  $T$  but converges more slowly to the true optimum as  $T$  grows, because the constraint remains binding even with accurate estimates. The minimum variance portfolio is competitive when  $T$  is small, benefiting from avoiding noisy mean estimates entirely. Notably, none of the strategies reach the true tangency Sharpe ratio (dashed blue line), highlighting the persistent cost of estimation uncertainty regardless of the approach taken.

The key practical lesson is that investors should prefer strategies that reduce estimation error directly, either by avoiding the noisiest inputs (equal weight, minimum variance), regularizing estimates (Ledoit-Wolf), or constraining the solution space (short-sell constraint) — rather than relying on unconstrained plug-in optimization with limited data.

## 8 Conclusion

Estimation error substantially reduces the out-of-sample performance of mean-variance optimal portfolios, especially when the sample size  $T$  is small relative to the number of assets  $N$ . The plug-in tangency portfolio can even produce negative Sharpe ratios at short horizons ( $T = 60$ ). Structured alternatives, naive diversification, minimum variance, constrained optimization, and covariance shrinkage, all mitigate this problem to varying degrees. Equal weighting is hardest to beat in short samples, while Ledoit-Wolf shrinkage offers the best risk-adjusted improvement among the parameter-based strategies. Overall, our analysis confirms that estimation risk is a first-order concern in portfolio construction and that simpler or regularized strategies offer meaningful practical benefits.

## 9 References

- Brandt, M. W. (2010). Chapter 5 - portfolio choice problems. In AÏT-SAHALIA, Y. and HANSEN, L. P., editors, *Handbook of Financial Econometrics: Tools and Techniques*, volume 1 of *Handbooks in Finance*, pages 269–336. North-Holland, San Diego.
- Jobson, J. D. and Korkie, B. (1980). Estimation for markowitz efficient portfolios. *Journal of the American Statistical Association*, 75(371):544–554.